UNIT 2 · TOPIC 12(A)

Drones — Types, Selection and Operation

Syllabus topic: Selection and operation of drones

Types

Platform	Strengths / limits
Multirotor (quad/hexa/octocopter)	Vertical take-off, hover, precise low-altitude spraying — the standard agri spray platform; limited endurance
Fixed-wing	Long endurance and large-area imaging/survey; needs launch/landing space; cannot hover
Hybrid VTOL	Combines both

Components

Brushless DC (BLDC) motors + electronic speed controllers (ESC); Li-polymer (LiPo) batteries; flight controller with IMU + GNSS; radio link; payloads — spray tank with pump and rotary-atomiser or hydraulic nozzles, spreader (granules/seed), cameras (RGB, multispectral for NDVI, thermal).

Selection criteria

Criterion	Typical value
Payload	10–16 L tank
Endurance	15–30 min per battery
Swath	4–6 m
Field capacity	≈ 1 ha per 10–15 min; 15–25 ha/day

Also weigh: atomiser type and droplet spectrum, terrain-follow radar, obstacle sensing, spares/service network, cost and subsidy eligibility.

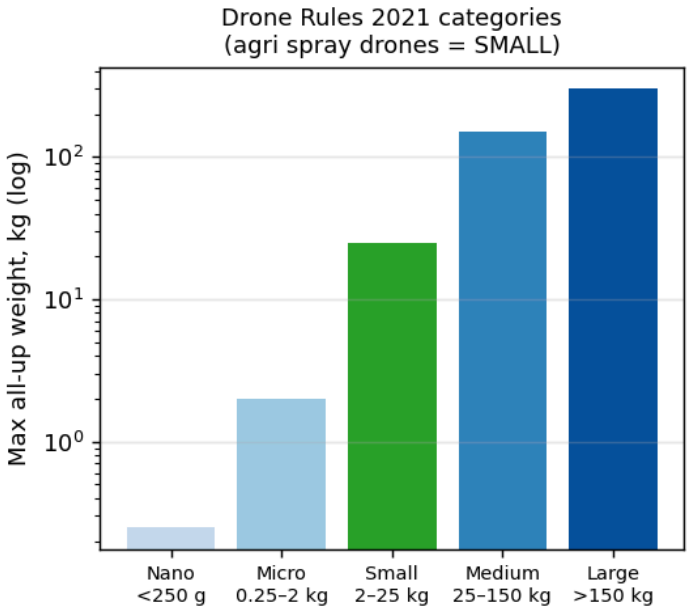
Spray operation (per MoA&FW SOP)

1. Check operator PPE and LiPo battery safety — mandatory checklist items.
2. Calibrate the flow to speed and swath.
3. Check the weather: do not spray in wind above **~10 km/h**, temperatures **>35 °C**, or rain.
4. Keep buffer zones near sensitive areas.
5. Set the spray volume, height and speed from the crop-specific SOP: about **20–25 L/ha** by crop stage (vs 400–500 L/ha with a knapsack), roughly **1.5–2.5 m** above the canopy (3–4 m in sugarcane), at about **3.5–5 m/s**.

The rotor downwash improves canopy penetration.

Figures reconciled (October 2026). The Unit 2 source gave 20–40 L/ha at 2–3 m and 3–6 m/s; Unit 1 gave 10–25 L/ha at 1.5–3 m. The government reference is the crop-specific SOP for applying pesticides with drones, released by MoA&FW on 20 April 2023 for 10 crops (rice, maize, cotton, groundnut, pigeon pea, safflower, sesame, soybean, sugarcane, wheat) and based on a reference drone with a 10-L tank weighing under 25 kg. The SOP fixes water

volume by crop and growth stage — its tables give 20 L/ha at the early stage, and summaries of the SOP put the range at 20–25 L/ha — and chooses height from drone weight and downwash, close to the canopy without lodging the crop. Always use the SOP table for the crop in hand.



Agri spray drone — typical operation

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Tank: 10–16 L
Swath: 4–6 m
Height above crop: 2–3 m
Speed: 3–6 m/s
Application rate: 20–40 L/ha
Field capacity: ~1 ha in 10–15 min
Avoid: wind > ~10 km/h,
  temp > 35 °C, rain
Battery: LiPo, 15–30 min/charge
Zones: fly in GREEN zone ≤120 m;
  UIN + Remote Pilot Certificate
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Drone Rules 2021 categories and typical spray-drone operation

Practice MCQs

Q209. Typical drone ULV spray application rate is:

(a) 2–4 L/ha
(b) 20–40 L/ha
(c) 200–400 L/ha
(d) 500–600 L/ha

Q210. Recommended spraying height of an agri drone above the crop canopy is about:

(a) 0.2–0.5 m
(b) 2–3 m
(c) 8–10 m
(d) 15–20 m

Q211. Drone spraying should be avoided when wind speed exceeds about:

(a) 2 km/h
(b) 10 km/h
(c) 30 km/h
(d) 50 km/h

Q212. Agricultural drones are commonly powered by:

(a) Lead-acid batteries
(b) LiPo (lithium-polymer) batteries
(c) NiCd
(d) Dry cells

Q215. For large-area aerial survey/imaging with long endurance, the preferred platform is:

(a) Quadcopter
(b) Fixed-wing drone
(c) Tethered balloon
(d) Hexacopter

Answers

Q209 — (b) 20–40 L/ha — *Closest option — the 2023 crop-specific SOP sets about 20–25 L/ha by crop stage (Topic 12a).*

Q210 — (b) 2–3 m — *Closest option — summaries of the 2023 SOP give about 1.5–2.5 m above the canopy (3–4 m in sugarcane).*

Q211 — (b) 10 km/h

Q212 — (b) LiPo (lithium-polymer) batteries

Q215 — (b) Fixed-wing drone

Sources: Kepner et al. (hydraulic principles); Drone Rules 2021 (Ministry of Civil Aviation); MoA&FW SOPs for drone application of pesticides & nutrients; SMAM drone guidelines & PIB releases (Kisan Drone, Namo Drone Didi — verified July 2026); ISO 11783 (ISOBUS); NPTEL lectures on mechatronics & precision agriculture.
